Preliminaries

Archer Yang

McGill University

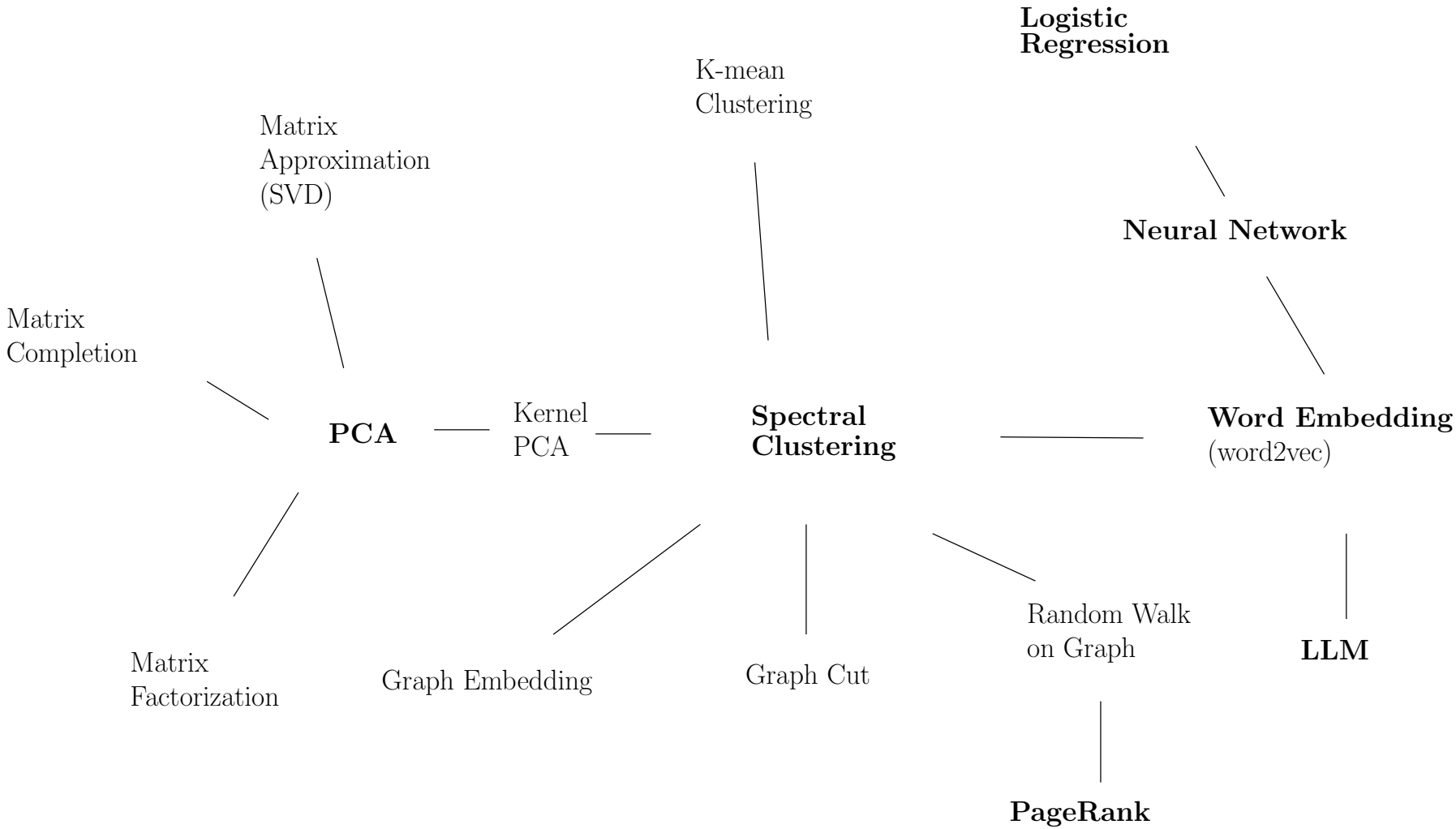
- **Reading assignment:** read chapter 3.1-3.8.2 of Mathematics for Machine Learning book.

We will learn

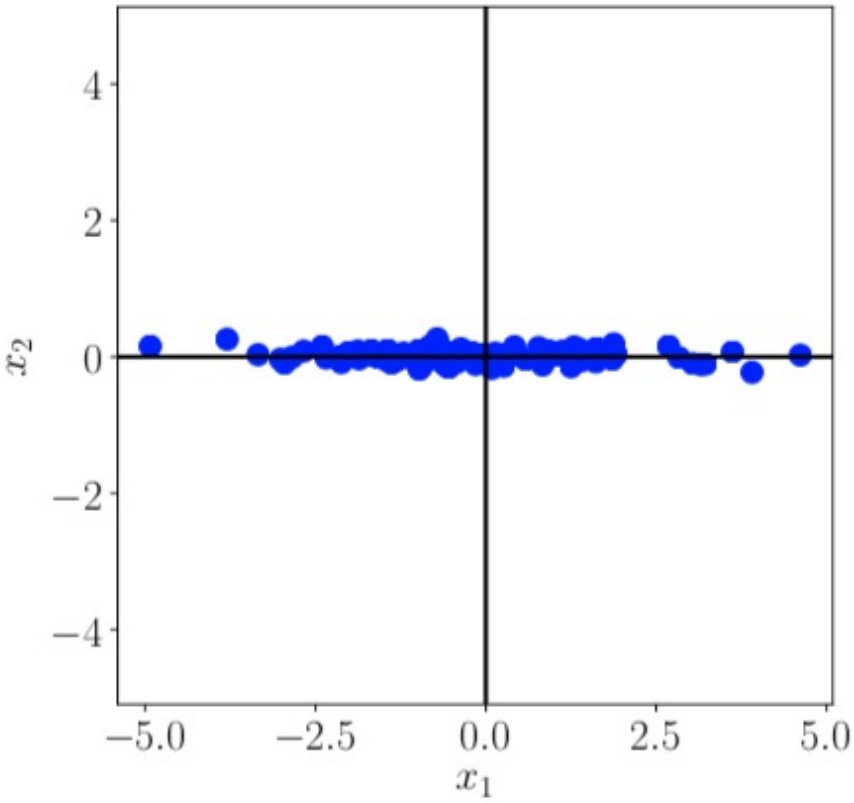
- How data (e.g., images) is compressed?
- The mechanism of online recommendation systems?
- How individuals can be clustered based on their genetic features?
- How Google ranks search results?
- How ML algorithms classify images?

Some implications

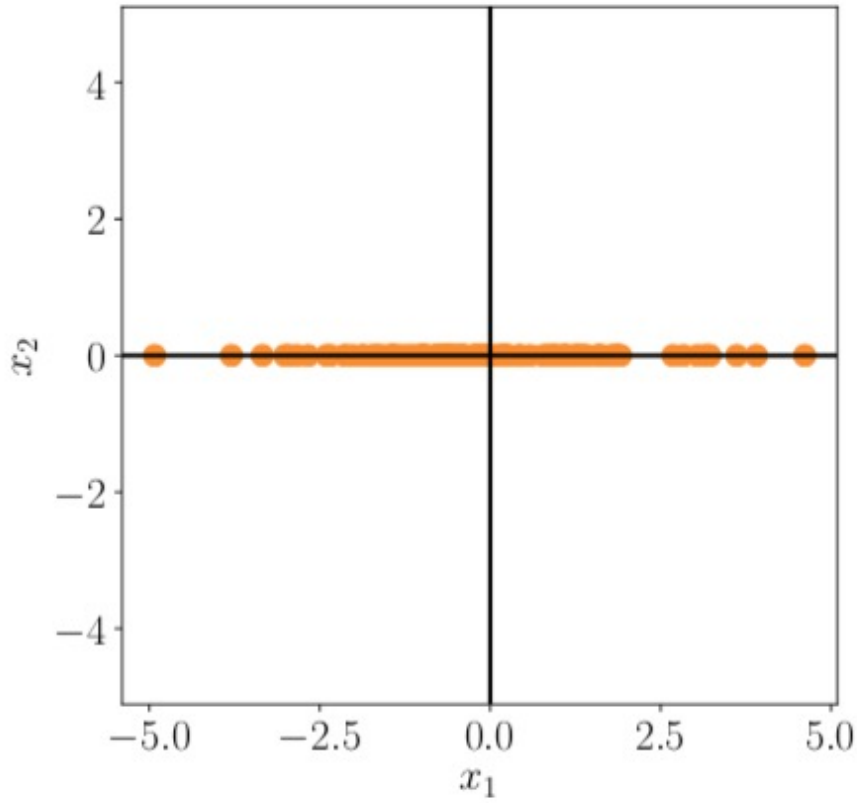
- How do we turn words into features for downstream tasks?
- How do ML models learn *abstract concepts*?
- Is a language model (AI) just a *language* model?
- Was AI invented or discovered?


Logistic Regression
K-mean Clustering
Matrix Approximation (SVD)
Neural Network
Matrix Completion
PCA
Kernel PCA
Spectral Clustering
Word Embedding (word2vec)
Random Walk on Graph
LLM
Matrix Factorization
Graph Embedding
Graph Cut
PageRank

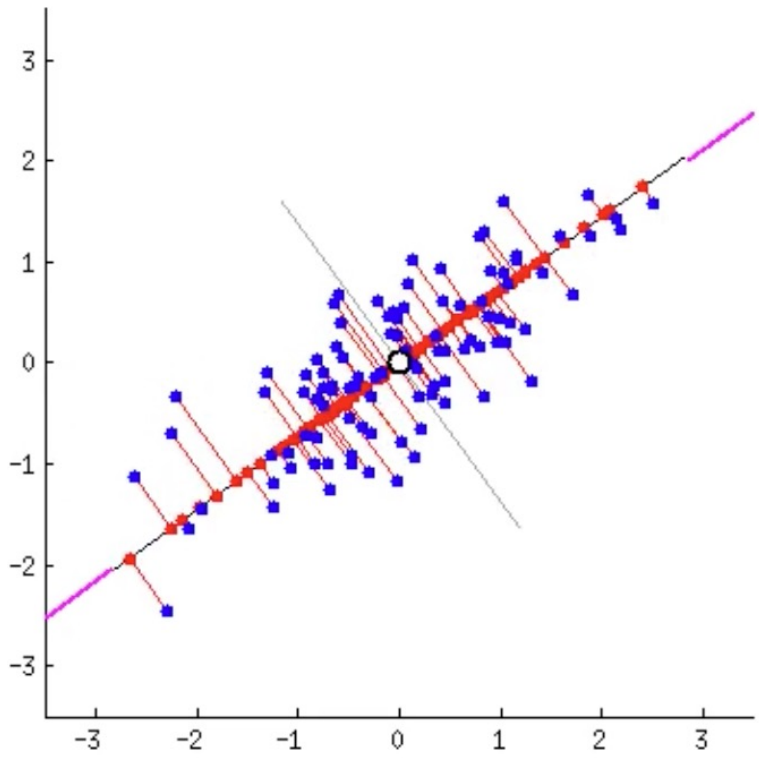
Intuition

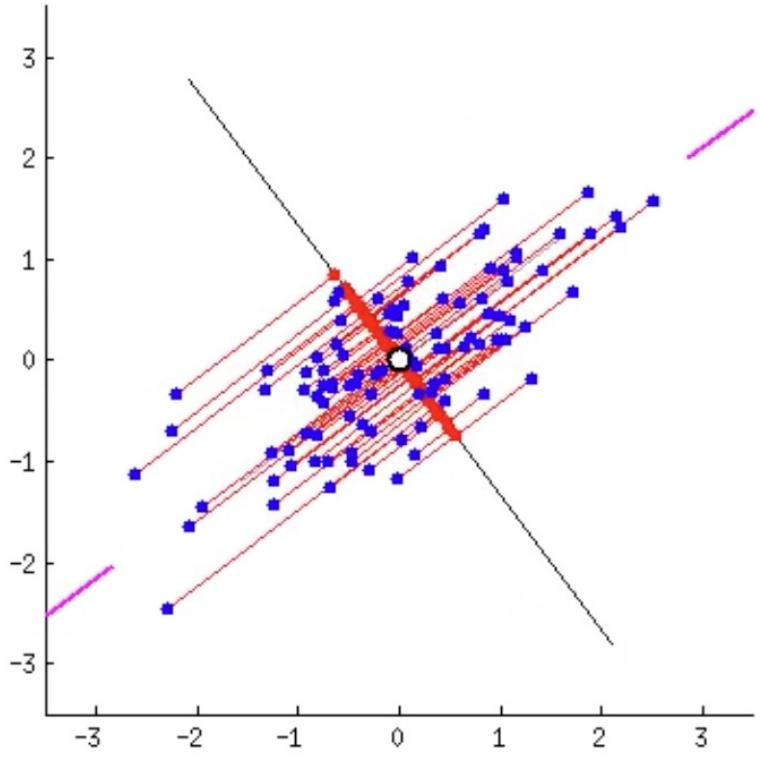


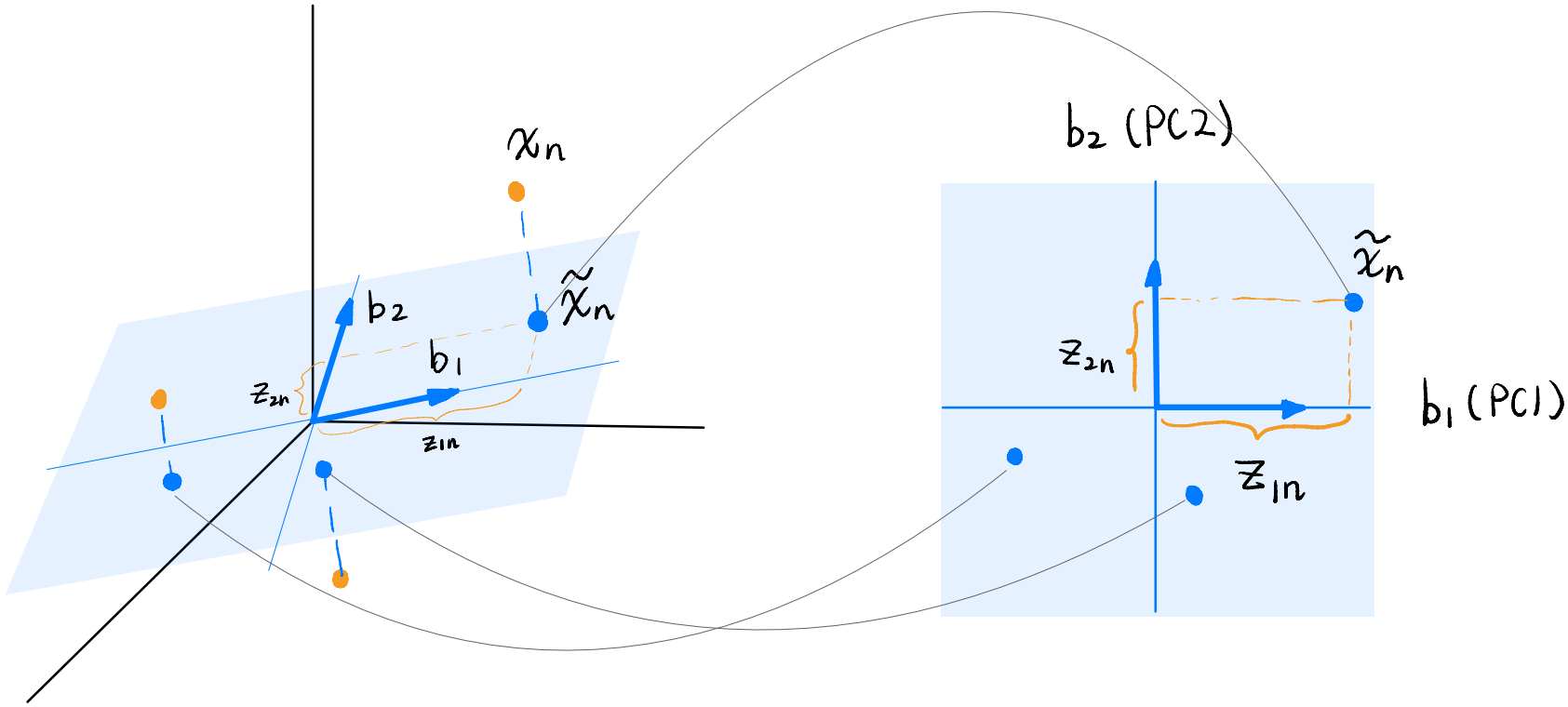
(a) Dataset with x_1 and x_2 coordinates.



(b) Compressed dataset where only the x_1 coordinate is relevant.


3
2
1
0
-1
-2
-3
-3
-2
-1
0
1
2
3

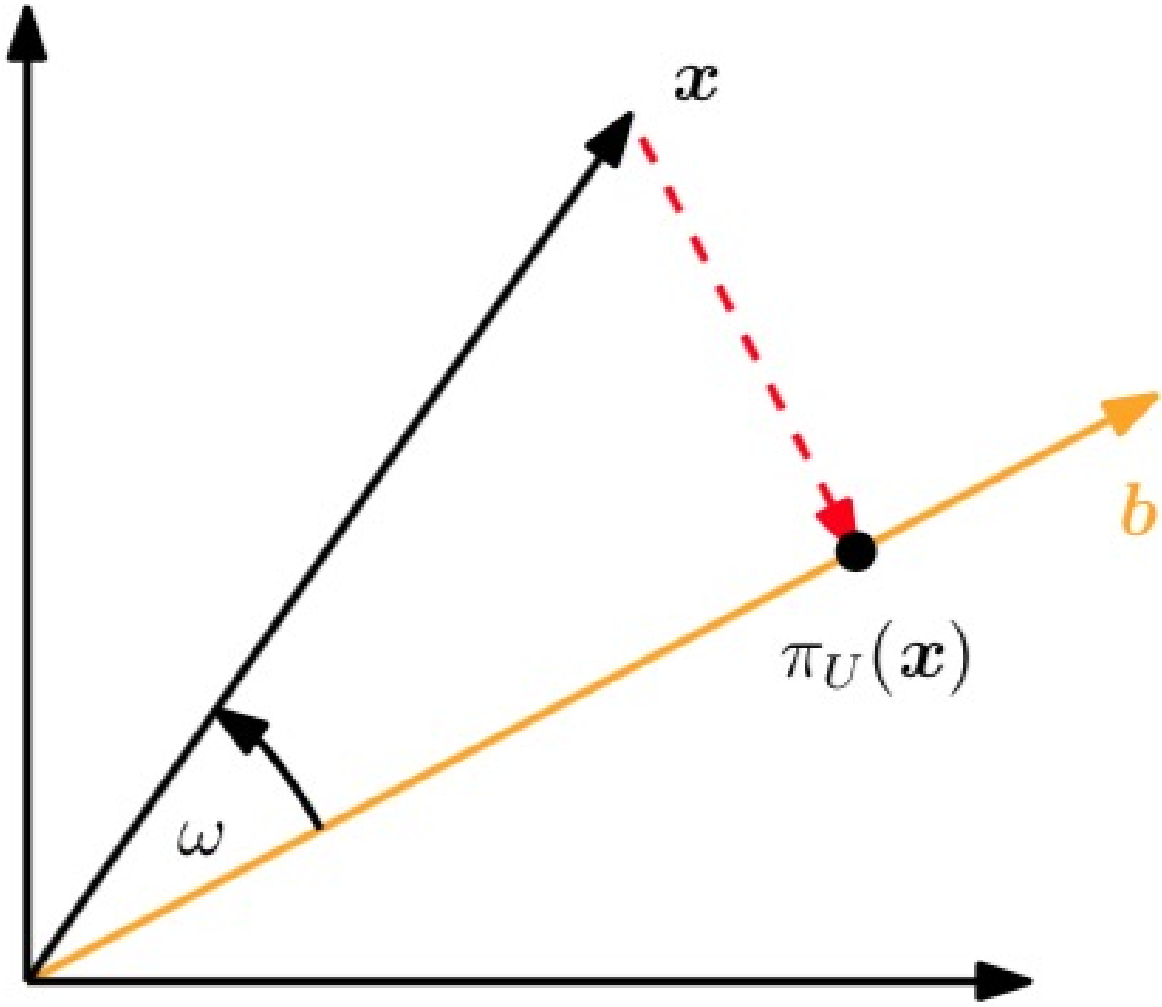

3
2
1
0
-1
-2
-3
-3
-2
-1
0
1
2
3


x_n
$\tilde{x}_n$
b_2
b_1
z_{2n}
z_{1n}
b_2 (PC2)
b_1 (PC1)

Orthogonal Projection onto 1D space

Orthogonal projection onto 1D subspaces

Task: project (orthogonally) vector $\boldsymbol{x}$ to a line U characterized by vector b, how to compute the projection vector $\pi_U(\boldsymbol{x})$?



Orthogonal projection onto 1D subspace

- Given a line through the origin with basis vector $\boldsymbol{b} = (b_1, \cdots, b_D)^\top \in \mathbb{R}^D$.
- Note that $\boldsymbol{b}$ is a **column vector**.
- The line through the origin creates a one-dimensional subspace U **spanned by** $\boldsymbol{b}$

$$U = \{z\boldsymbol{b} : \forall z \in \mathbb{R}\}$$

- Now we project $\boldsymbol{x} = (x_1, \cdots, x_D)^\top \in \mathbb{R}^D$ onto U, denote this **projection** as $\pi_U(\boldsymbol{x})$

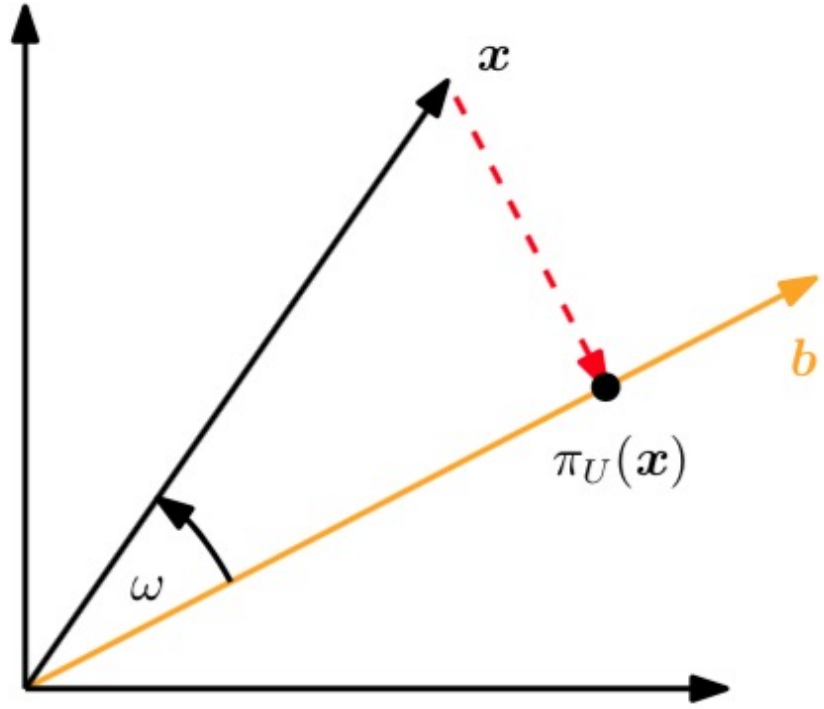
Orthogonal projection onto 1D subspace

- The projection $\pi_U(\boldsymbol{x})$ of $\boldsymbol{x}$ onto U **must be an element** of U, i.e.

$$\pi_U(\boldsymbol{x}) = z\boldsymbol{b}$$

  for **some** $z \in \mathbb{R}$.

- How to determine the coordinate z?



Finding the projection point

To solve coordinate z:

1. The orthogonality condition yields

$$\langle \boldsymbol{x} - \pi_U(\boldsymbol{x}), \boldsymbol{b} \rangle = 0 \overset{\pi_U(\boldsymbol{x}) = z\boldsymbol{b}}{\iff} \langle \boldsymbol{x} - z\boldsymbol{b}, \boldsymbol{b} \rangle = 0.$$

2. By the bilinearity of the inner product

$$\langle \boldsymbol{x}, \boldsymbol{b} \rangle - z\langle \boldsymbol{b}, \boldsymbol{b} \rangle = 0 \iff z = \frac{\langle \boldsymbol{x}, \boldsymbol{b} \rangle}{\langle \boldsymbol{b}, \boldsymbol{b} \rangle} = \frac{\langle \boldsymbol{b}, \boldsymbol{x} \rangle}{\|\boldsymbol{b}\|^2}$$

3. In a Euclidean space, $\langle \cdot, \cdot \rangle$ is the dot product

$$z = \frac{\boldsymbol{b}^\top \boldsymbol{x}}{\boldsymbol{b}^\top \boldsymbol{b}} = \frac{\boldsymbol{b}^\top \boldsymbol{x}}{\|\boldsymbol{b}\|^2}$$

Since $\pi_U(\boldsymbol{x}) = z\boldsymbol{b}$,

$$\pi_U(\boldsymbol{x}) = z\boldsymbol{b} = \frac{\boldsymbol{b}^\top \boldsymbol{x}}{\|\boldsymbol{b}\|^2}\boldsymbol{b}$$

- z is the coordinate of projection, since the length of $\pi_U(\boldsymbol{x})$ is

$$\|\pi_U(\boldsymbol{x})\| = \|z\boldsymbol{b}\| = |z|\|\boldsymbol{b}\|$$

Special case $\|\boldsymbol{b}\| = 1$

- Special case: if $\|\boldsymbol{b}\| = 1$, the **coordinate** z of the projection is

$$z = \frac{\boldsymbol{b}^\top \boldsymbol{x}}{\|\boldsymbol{b}\|^2} = \boldsymbol{b}^\top \boldsymbol{x}$$

  The projection is $\pi_U(\boldsymbol{x})$

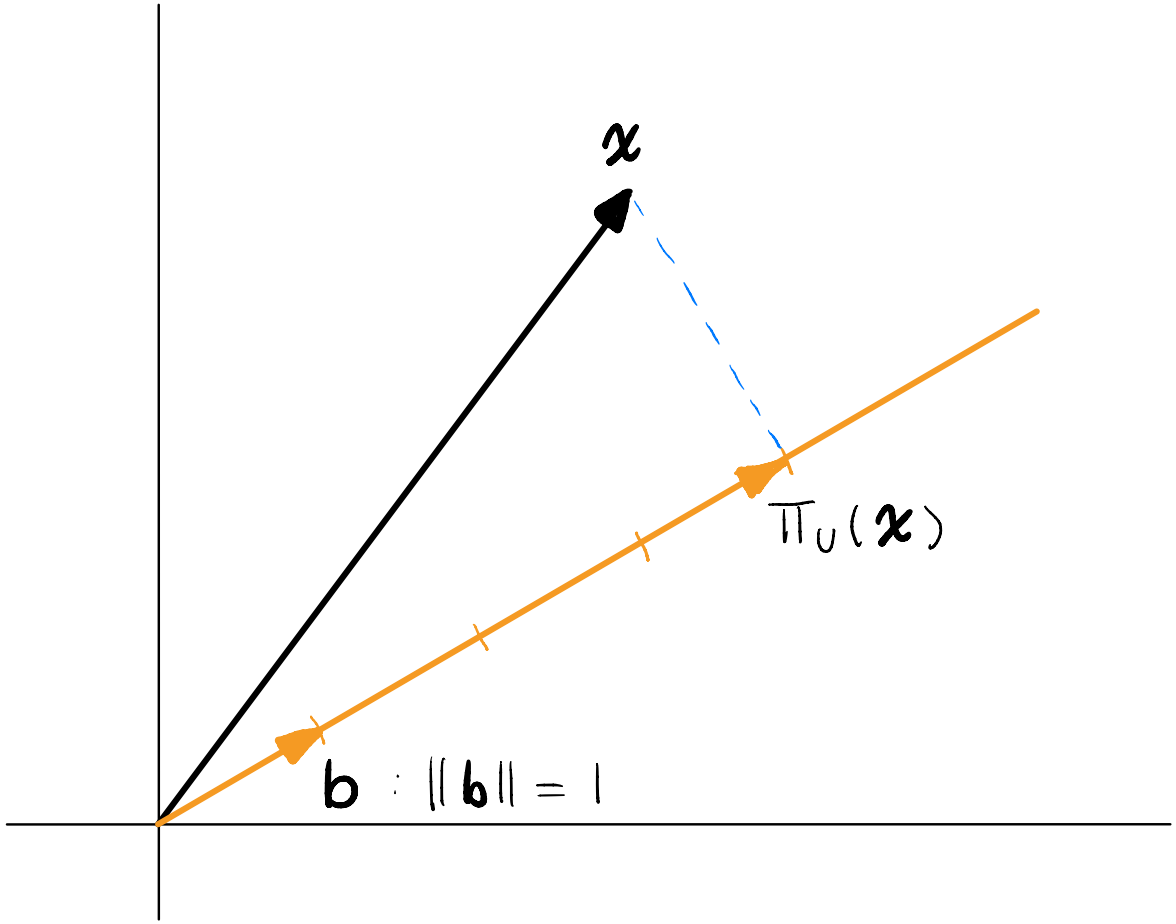
$$\pi_U(\boldsymbol{x}) = z\boldsymbol{b} = \boldsymbol{b}\boldsymbol{b}^\top \boldsymbol{x}$$

- The length of $\pi_U(\boldsymbol{x})$ is

$$\|\pi_U(\boldsymbol{x})\| = \|z\boldsymbol{b}\| = |z|\|\boldsymbol{b}\| = |z|$$

Visualization

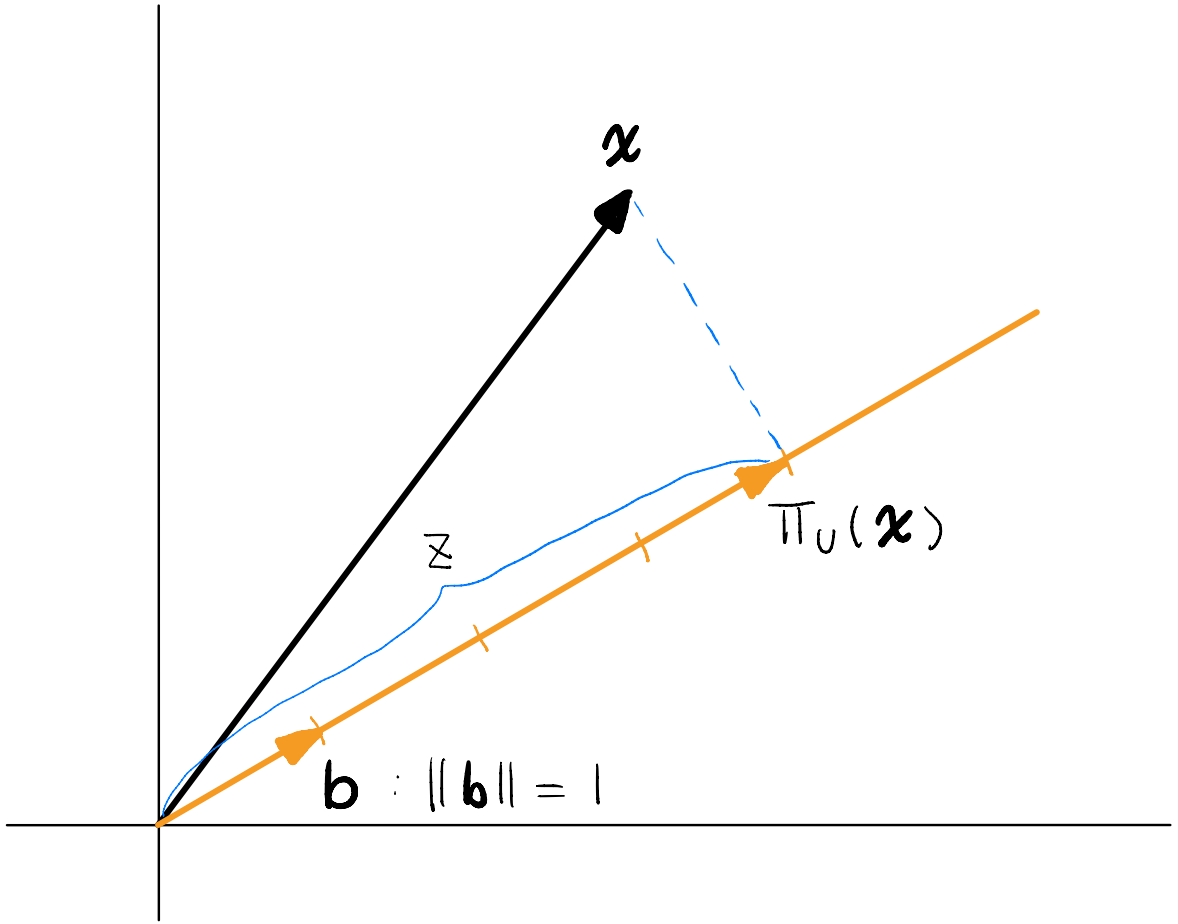
When $||b|| = 1$



Visualization

When $\|\boldsymbol{b}\| = 1$, the **coordinate** z of the projection is

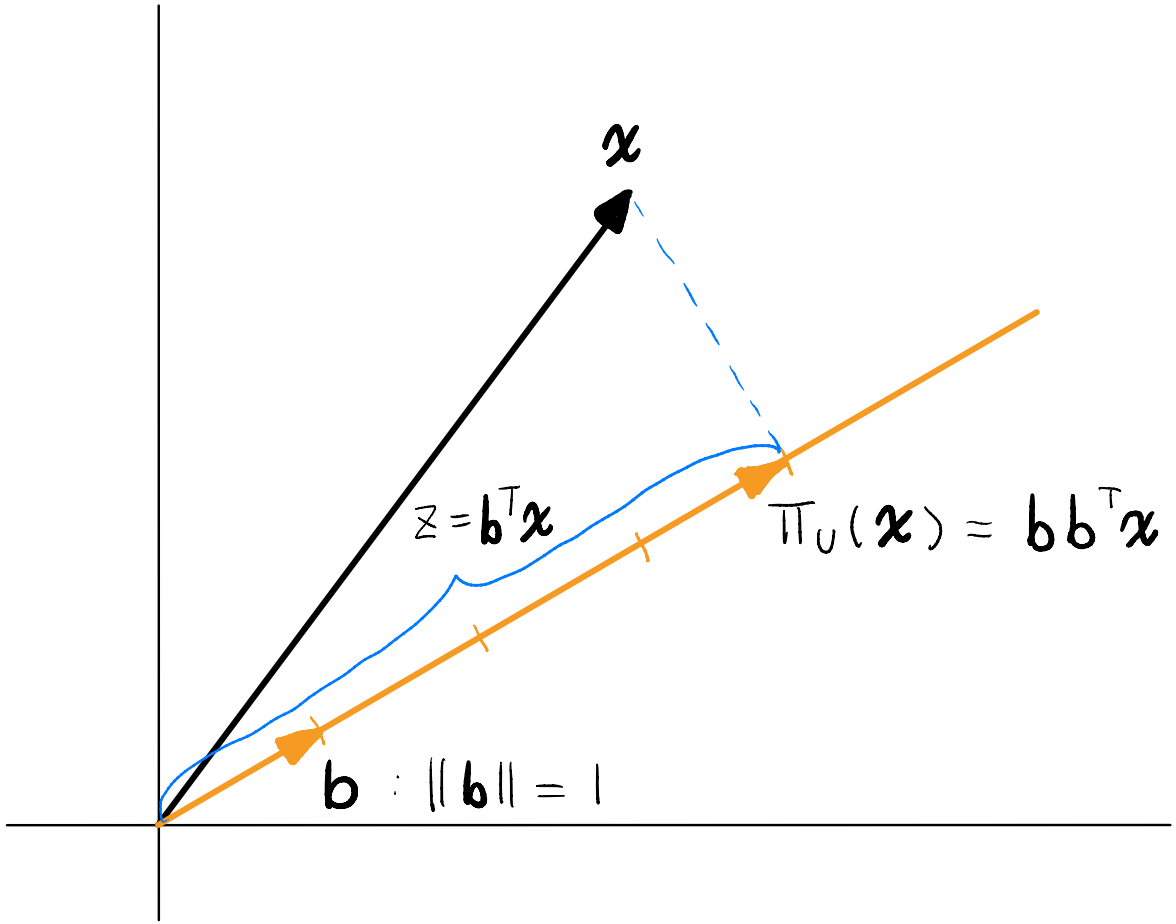
$$z = \boldsymbol{b}^\top \boldsymbol{x} = \|\boldsymbol{x}\|\|\boldsymbol{b}\| \cos\omega = \|\boldsymbol{x}\| \cos\omega$$



Visualization

When $||b|| = 1$, the **projection** $\pi_U(x)$ is

$$\pi_U(x) = zb = bz = bb^\top x$$



A practice

- (Homework) Given that

$$\boldsymbol{x}^\top \boldsymbol{y} = \|\boldsymbol{x}\| \|\boldsymbol{y}\| \cos\omega$$

where ω is the angle between $\boldsymbol{x}$ and $\boldsymbol{y}$. Show that

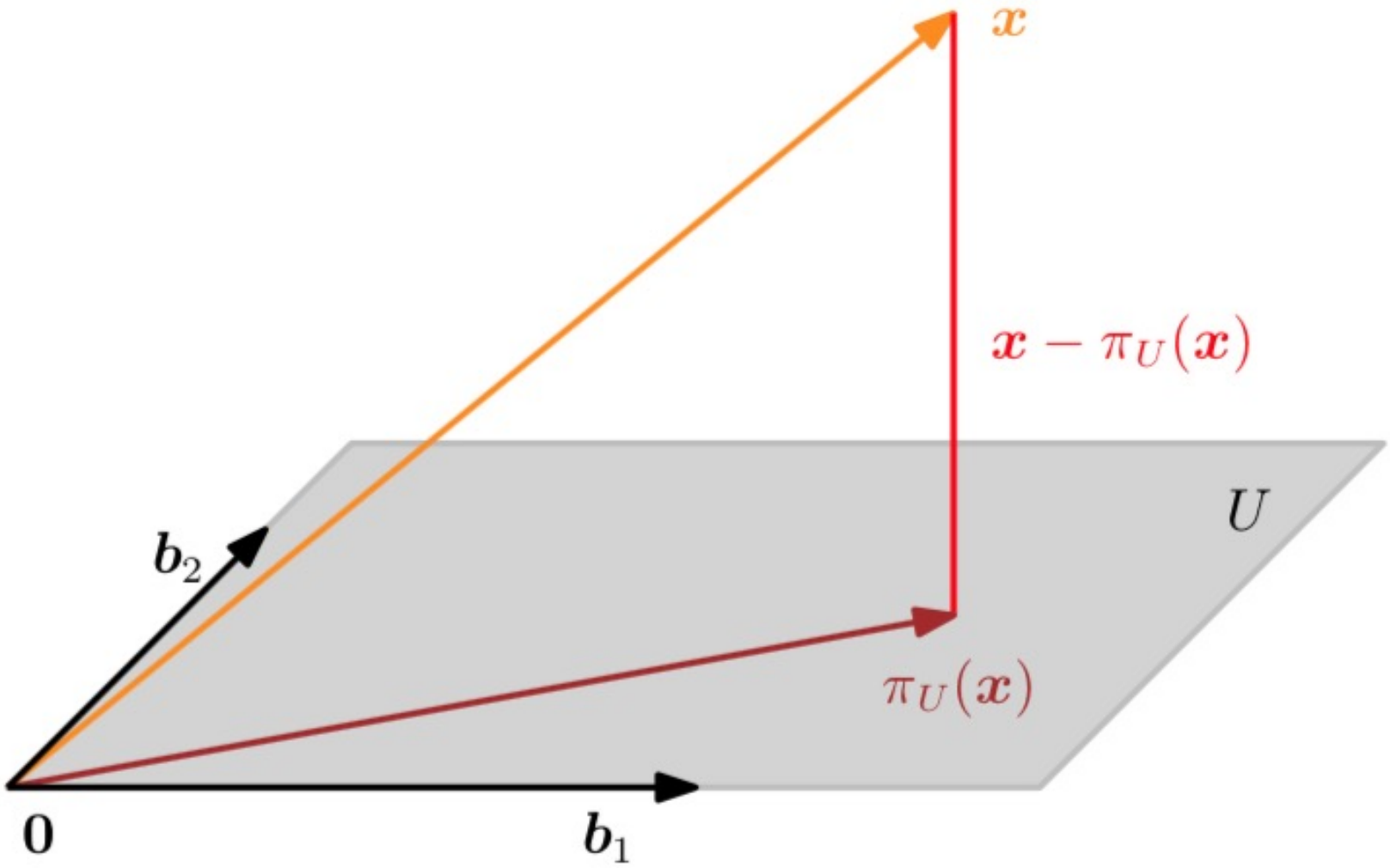
$$\|\pi_U(\boldsymbol{x})\| = |\cos\omega| \|\boldsymbol{x}\|$$

- (Homework) If $\|\boldsymbol{x}\| = 1$ (lies on the unit circle), then $\|\pi_U(\boldsymbol{x})\| = |\cos\omega|$

Orthogonal Projection onto General Subspaces

Orthogonal projection onto general subspaces

Now consider orthogonal projections of $\boldsymbol{x} \in \mathbb{R}^D$ onto lower-dimensional subspaces $U \subseteq \mathbb{R}^D$ with $\dim(U) = M \geq 1$.



Orthogonal projection onto general subspaces

Assume $\boldsymbol{b}_1, \cdots, \boldsymbol{b}_M$ is an ordered basis of U

- Projection $\pi_U(\boldsymbol{x})$ **must be an element** of U, i.e. $\pi_U(\boldsymbol{x})$ can be represented as linear combinations of the basis vectors

$$\boldsymbol{b}_1, \cdots, \boldsymbol{b}_M$$

of U

$$\pi_U(\boldsymbol{x}) = \sum_{i=1}^{M} z_i \boldsymbol{b}_i = z_1 \boldsymbol{b}_1 + \cdots + z_M \boldsymbol{b}_M$$

- How to determine $z_1, \cdots, z_M$?

Finding the coordinates $z_1, \cdots, z_M$

- First we use the matrix notation

$$\pi_U(\boldsymbol{x}) = \sum_{i=1}^{M} z_i \boldsymbol{b}_i = \boldsymbol{B}\boldsymbol{z}$$

where

$$\boldsymbol{B} = \begin{pmatrix} | & | & & | \\ \boldsymbol{b}_1 & \boldsymbol{b}_2 & \cdots & \boldsymbol{b}_M \\ | & | & & | \end{pmatrix} \in \mathbb{R}^{D \times M}, \qquad \boldsymbol{z} = (z_1, \cdots, z_M)^\top \in \mathbb{R}^M$$

Finding the coordinates $z_1, \cdots, z_M$

- Need to find $\pi_U(\boldsymbol{x})$ in U that is closest to $\boldsymbol{x} \in \mathbb{R}^D$.
- This implies that $\boldsymbol{x} - \pi_U(\boldsymbol{x})$ must be orthogonal to all basis vectors of U:

$$\langle \boldsymbol{b}_1, \boldsymbol{x} - \pi_U(\boldsymbol{x}) \rangle = \boldsymbol{b}_1^\top (\boldsymbol{x} - \pi_U(\boldsymbol{x})) = 0$$

$$\vdots$$

$$\langle \boldsymbol{b}_M, \boldsymbol{x} - \pi_U(\boldsymbol{x}) \rangle = \boldsymbol{b}_M^\top (\boldsymbol{x} - \pi_U(\boldsymbol{x})) = 0$$

Finding the coordinates $z_1, \cdots, z_M$

- With $\pi_U(\boldsymbol{x}) = \boldsymbol{B}\boldsymbol{z}$

$$\boldsymbol{b}_1^\top (\boldsymbol{x} - \boldsymbol{B}\boldsymbol{z}) = 0$$

$$\vdots$$

$$\boldsymbol{b}_M^\top (\boldsymbol{x} - \boldsymbol{B}\boldsymbol{z}) = 0$$

- In a matrix format

$$\begin{bmatrix} \boldsymbol{b}_1^\top \\ \vdots \\ \boldsymbol{b}_M^\top \end{bmatrix} [\boldsymbol{x} - \boldsymbol{B}\boldsymbol{z}] = \boldsymbol{0} \iff \boldsymbol{B}^\top (\boldsymbol{x} - \boldsymbol{B}\boldsymbol{z}) = \boldsymbol{0}$$

$$\iff \boldsymbol{B}^\top \boldsymbol{B}\boldsymbol{z} = \boldsymbol{B}^\top \boldsymbol{x}.$$

- We get the coordinate of the orthogonal projection

$$\boldsymbol{z} = (\boldsymbol{B}^\top \boldsymbol{B})^{-1} \boldsymbol{B}^\top \boldsymbol{x}$$

Finding the projection point

- We already know that

$$\pi_U(\boldsymbol{x}) = \boldsymbol{B}\boldsymbol{z}$$

- Therefore the **projection** $\pi_U(\boldsymbol{x})$ is

$$\pi_U(\boldsymbol{x}) = \boldsymbol{B}(\boldsymbol{B}^\top \boldsymbol{B})^{-1}\boldsymbol{B}^\top \boldsymbol{x}$$

$B^\top B$ is invertible

- Since $\boldsymbol{b}_1, \cdots, \boldsymbol{b}_M$ are a basis of U, they are linearly independent, so

$$\text{rank}(\boldsymbol{B}) = M \qquad \text{where } M \leq D$$

- $\boldsymbol{B}^\top \boldsymbol{B} \in \mathbb{R}^{M \times M}$ is invertible since

$$\text{rank}(\boldsymbol{B}^\top \boldsymbol{B}) = \text{rank}(\boldsymbol{B}) = M \tag{1}$$

Proof of Eq. (1)

- First we see that nullity$(\boldsymbol{B}^\top \boldsymbol{B})$ = nullity$(\boldsymbol{B})$, this is because for any $\boldsymbol{x}$ that satisfies

$$\boldsymbol{B}^\top \boldsymbol{B}\boldsymbol{x} = \boldsymbol{0}$$

we also have

$$\|\boldsymbol{B}\boldsymbol{x}\|^2 = \boldsymbol{x}\boldsymbol{B}^\top \boldsymbol{B}\boldsymbol{x} = 0$$

hence $\boldsymbol{B}\boldsymbol{x} = \boldsymbol{0}$. Therefore nullity$(\boldsymbol{B}^\top \boldsymbol{B})$ = nullity$(\boldsymbol{B})$

- By rank–nullity theorem, we have nullity$(\boldsymbol{B})$ = M − rank$(\boldsymbol{B})$ = 0, then

$$\text{rank}(\boldsymbol{B}^\top \boldsymbol{B}) = M - \text{nullity}(\boldsymbol{B}^\top \boldsymbol{B}) = M - \text{nullity}(\boldsymbol{B}) = M$$

Remark:

- The projections $\pi_U(\boldsymbol{x})$ are still vectors in $\mathbb{R}^D$ although they lie in an M-dimensional subspace $U \subseteq \mathbb{R}^D$.
- However, to represent a projected vector we only need the M coordinates $z_1, \cdots, z_M$ ($M < D$) with respect to the basis vectors $\boldsymbol{b}_1, \cdots, \boldsymbol{b}_M$.

Special case: orthonormal basis

- If the basis $\{\boldsymbol{b}_1, \cdots, \boldsymbol{b}_M\}$ for the subspace U is an orthonomal basis, i.e for all $i, j = 1, \cdots, M$

$$\langle \boldsymbol{b}_i, \boldsymbol{b}_j \rangle = \boldsymbol{b}_i^\top \boldsymbol{b}_j = 0 \qquad \text{for } i \neq j$$

$$\langle \boldsymbol{b}_i, \boldsymbol{b}_i \rangle = \boldsymbol{b}_i^\top \boldsymbol{b}_i = 1$$

Then

$$\boldsymbol{B}^\top \boldsymbol{B} = (\boldsymbol{b}_1, \cdots, \boldsymbol{b}_M)^\top (\boldsymbol{b}_1, \cdots, \boldsymbol{b}_M) = \boldsymbol{I}_{M \times M}$$

Special case: orthonormal basis

- In this case, the coordinate of the orthogonal projection reduces to

$$\boldsymbol{z} = (\boldsymbol{B}^\top \boldsymbol{B})^{-1} \boldsymbol{B}^\top \boldsymbol{x} = \boldsymbol{I}_{M\times M}^{-1} \boldsymbol{B}^\top \boldsymbol{x} = \boldsymbol{B}^\top \boldsymbol{x}$$

  and the **projection** $\pi_U(\boldsymbol{x})$ reduces to

$$\pi_U(\boldsymbol{x}) = \boldsymbol{B}\boldsymbol{z} = \boldsymbol{B}\boldsymbol{B}^\top \boldsymbol{x}$$

Sample Mean and Sample Variance

Sample mean and sample variance

- Given N observations of a random variable X, denoted as $x_1, x_2, \cdots, x_N$.
- Sample mean - captures the center of $x_1, x_2, \cdots, x_N$,

$$\bar{x} = \frac{1}{N} \sum_{n=1}^{N} x_n$$

- Sample variance – captures the variation of $x_1, x_2, \cdots, x_N$,

$$s_X^2 \equiv \widehat{\mathrm{Var}}(X) = \frac{1}{N-1} \sum_{n=1}^{N} (x_n - \bar{x})^2$$

SAMPLE COVARIANCE

- Given N observations of two variables X and Y,

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} \quad \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}$$

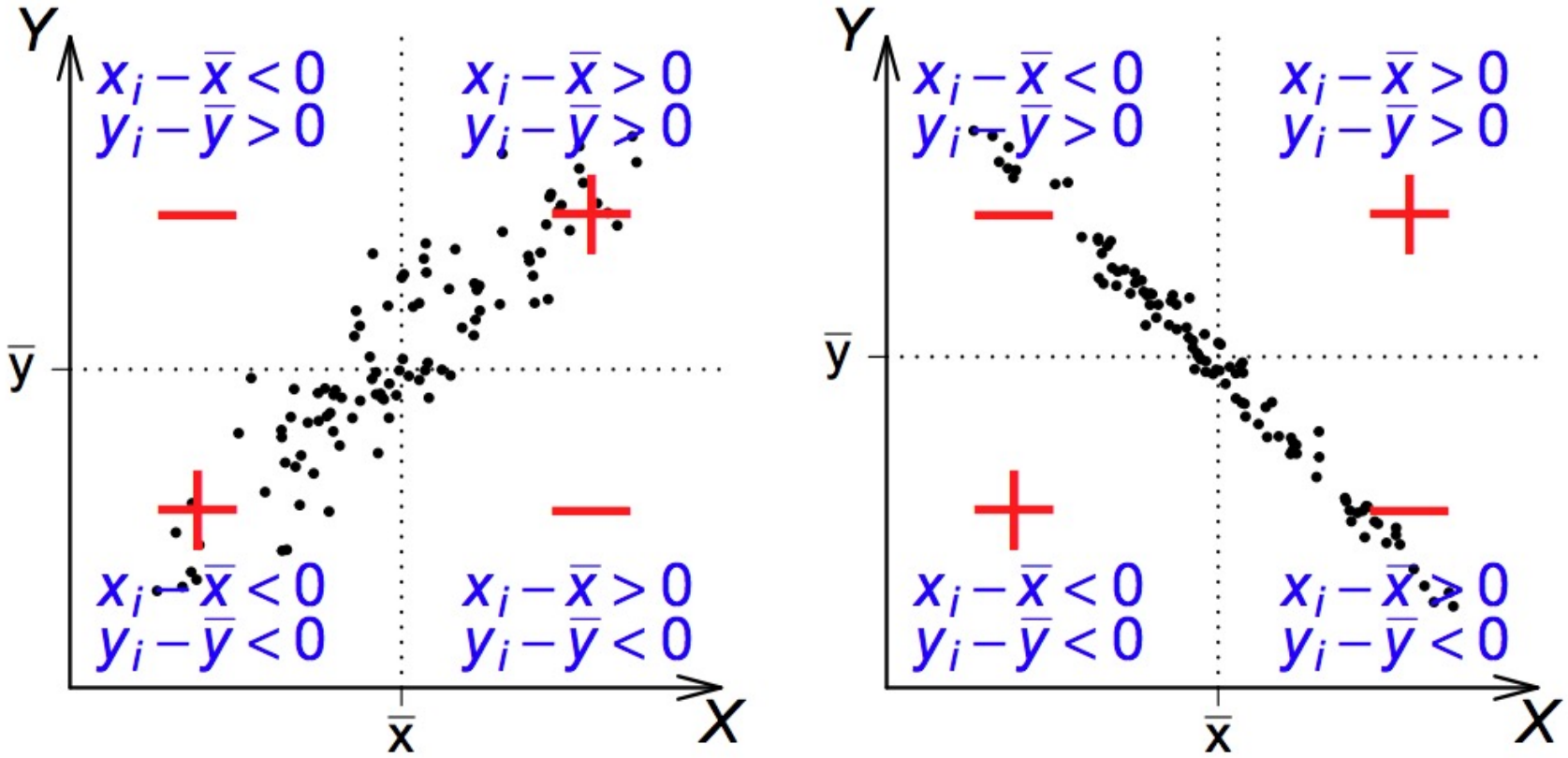
- Sample covariance s_{XY} is a measure of the **direction** and **strength** of the linear relationship between X and Y, defined as

$$s_{XY} = \frac{1}{N-1} \sum_{n=1}^{N} (x_n - \bar{x})(y_n - \bar{y})$$

- $s_{XY} > 0$ – positive linear relation
- $s_{XY} < 0$ – negative linear relation
- The **magnitude** of covariance reflects the **strength** of the relation

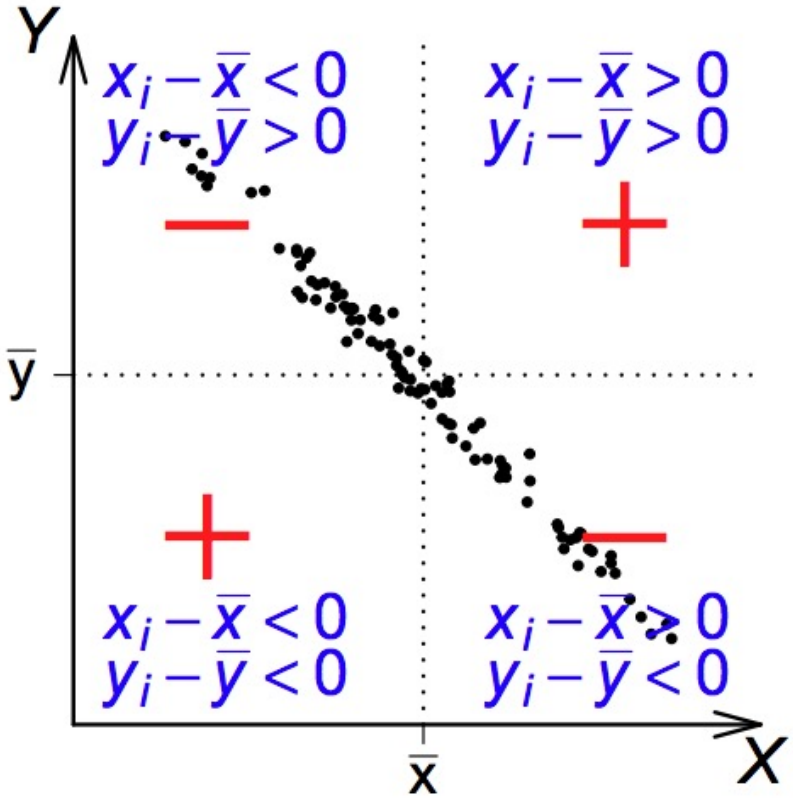
Interpretation

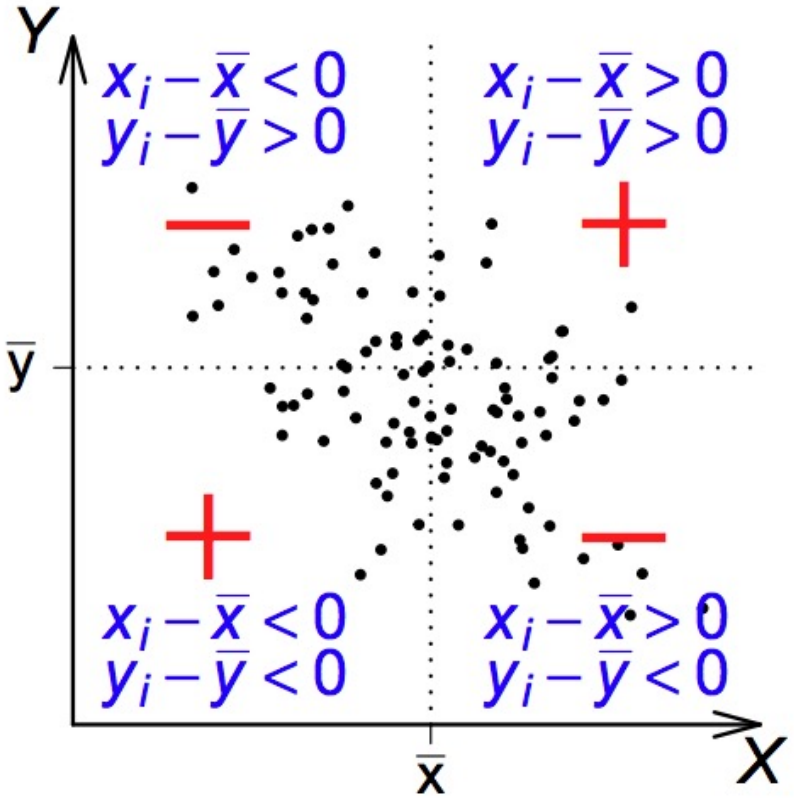
- The **sign** of s_{XY} reflects the **direction** of the linear relation



Interpretation

- The **magnitude** of s_{XY} reflects the **strength** of the linear relation





Sample Covariance Matrix

- Now suppose that there are p variables $X_1, X_2, \cdots, X_p$, each variable has N observations

$$\begin{bmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{N1} \end{bmatrix} \begin{bmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{N2} \end{bmatrix} \cdots \begin{bmatrix} x_{1p} \\ x_{2p} \\ \vdots \\ x_{Np} \end{bmatrix}$$

- Can be written in a matrix format

$$\boldsymbol{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & & \vdots \\ x_{N1} & x_{N2} & \cdots & x_{Np} \end{bmatrix}$$

Sample covariance matrix

- What is the sample covariance between variable X_j and X_k?
- When $j \neq k$

$$s_{jk} = s_{X_j X_k} = \frac{1}{N-1} \sum_{n=1}^{N} (x_{nj} - \bar{x}_j)(x_{nk} - \bar{x}_k)$$

- When $j = k$

$$s_{jj} = s_{X_j X_j} = \frac{1}{N-1} \sum_{n=1}^{N} (x_{nj} - \bar{x}_j)^2$$

Sample covariance matrix

- The sample covariance matrix is defined as

$$S = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1p} \\ s_{21} & s_{22} & \cdots & s_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s_{p1} & s_{p2} & \cdots & s_{pp} \end{bmatrix}$$

Sample covariance matrix

- Then

$$\boldsymbol{S} = \begin{bmatrix} \frac{1}{N-1}\sum_{n=1}^{N}(x_{n1}-\bar{x}_1)^2 & \cdots & \frac{1}{N-1}\sum_{n=1}^{N}(x_{n1}-\bar{x}_1)(x_{np}-\bar{x}_p) \\ \frac{1}{N-1}\sum_{n=1}^{N}(x_{n2}-\bar{x}_2)(x_{n1}-\bar{x}_1) & \cdots & \frac{1}{N-1}\sum_{n=1}^{N}(x_{n2}-\bar{x}_2)(x_{np}-\bar{x}_p) \\ \vdots & \ddots & \vdots \\ \frac{1}{N-1}\sum_{n=1}^{N}(x_{np}-\bar{x}_p)(x_{n1}-\bar{x}_1) & \cdots & \frac{1}{N-1}\sum_{n=1}^{N}(x_{np}-\bar{x}_p)^2 \end{bmatrix}$$

Sample covariance matrix

- Notice that if the data has already been centered

$$\hat{\boldsymbol{\mu}} = (\bar{x}_1, \cdots, \bar{x}_p)^\top = (0, \cdots, 0)^\top$$

- Then

$$\boldsymbol{S} = \begin{bmatrix} \frac{1}{N-1}\sum_{n=1}^N x_{n1}^2 & \frac{1}{N-1}\sum_{n=1}^N x_{n1}x_{n2} & \cdots & \frac{1}{N-1}\sum_{n=1}^N x_{n1}x_{np} \\ \frac{1}{N-1}\sum_{n=1}^N x_{n2}x_{n1} & \frac{1}{N-1}\sum_{n=1}^N x_{n2}^2 & \cdots & \frac{1}{N-1}\sum_{n=1}^N x_{n2}x_{np} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{N-1}\sum_{n=1}^N x_{np}x_{n1} & \frac{1}{N-1}\sum_{n=1}^N x_{np}x_{n2} & \cdots & \frac{1}{N-1}\sum_{n=1}^N x_{np}^2 \end{bmatrix}$$

- You can check that

$$\boldsymbol{S} = \frac{1}{N-1}\sum_{n=1}^N \boldsymbol{x}_n \boldsymbol{x}_n^\top = \frac{1}{N-1}\boldsymbol{X}^\top \boldsymbol{X}$$

where $\boldsymbol{x}_n = (x_{n1}, \cdots, x_{np})^\top$ for $n = 1, \cdots, N$.